

APPOINTMENTS

Columbia University in the City of New York NASA Hubble Fellowship Program Sagan Fellow	New York, New York Aug 2024–Aug 2027
University of Nevada, Las Vegas Graduate Research / Teaching Assistant	Las Vegas, Nevada Aug 2018–May 2024

EDUCATION

University of Nevada, Las Vegas Ph.D. in Astronomy, Advisor: Prof. Zhaohuan Zhu	Las Vegas, Nevada Sep 2018–May 2024
University of Michigan, Ann Arbor B.S. in Astronomy and Physics, Advisor: Prof. Lee Hartmann – GPA: 4.00 / 4.00, with Highest Distinction	Ann Arbor, Michigan 2016–2018
Nanjing University B.S. in Astronomy – GPA: 4.61 / 5.00, Rank 1 / 46	Nanjing, China 2014–2016

FELLOWSHIPS AND GRANTS

• Flatiron Institute Research Fellow , Center for Computational Astrophysics <i>Incoming</i>	2027–2028
• NASA Hubble Fellowship Program (NHFP) Sagan Fellow <i>~320,000 USD + 150,000 SBU supercomputer node hours per year</i>	2024–2027
• NASA FINESST Fellowship <i>Future Investigators in NASA Earth and Space Science and Technology; 135,000 USD + 200,000 SBU supercomputer node hours per year</i>	2021–2024
• UNLV Russell L. and Brenda Frank Scholarship (7,000 USD)	2022–2024
• AAS International Travel Grant (1,700 USD)	2023
• UNLV GPSA Travel Fund (5,000 USD)	2023
• UNLV Barrick Graduate Fellowship (30,000 USD)	2020–2021

OBSERVING PROGRAMS

• Principal Investigator , ALMA Cycle 11 (2024.1.00581.S) <i>Probing the radial dependence of midplane turbulence in a seven-ringed disk</i>	2024
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- **Team Member**, ALMA Cycle 11 Large Program [CHEER](#) (2024.1.01001.L) 2026–Present
Chemistry in Herbig Environments and their Exoplanet Relationships; constraining cooling timescales and shadow–disk interactions
- **Team Member**, ALMA Cycle 4 Large Program [DSHARP](#) (2016.1.00484.L) 2018
Disk Substructures at High Angular Resolution Project; led the planet–disk interaction interpretation paper (DSHARP VII)
- **Co-Investigator**, 11 additional ALMA programs, Cycles 7–13 2019–2026
PIs: J. Huang, F. Long, L. Pérez, A. Isella, N. Kurtovic, K. Mauco, S. Jiang
- **Co-Investigator**, 2 JWST GO programs Cycles 1, 4
GO 2540 (PI L. Ricci), disk–planet interaction in HD 163296; GO 7993 (PI J. Huang), disk substructures, water enrichment and pebble drift
- **Co-Investigator**, 2 VLT/MUSE programs, ESO Period 117 2026
117.29Z6 (PI N. Bajaj), jets tracing inner disk dynamics; 117.2B05 (PI C. Manara), MUSE-NFM of JWST-identified proplyds in σ Orionis

HONORS AND AWARDS

- [2024 IAU Ph.D. Prize, Division F — Honourable Mention](#) 2025
- [UNLV Outstanding Spring 2024 Graduate](#) 2024
Recognized by the university president at commencement as the sole Ph.D. honoree
- **OISS Distinguished Contribution Award** (1,000 USD) 2023
- **Bronze Medal**, 7th International Olympiad on Astronomy and Astrophysics Volos, Greece, 2013

PUBLICATIONS

All papers (35 of 37 [on ADS](#)): 4,398 citations, h-index 21. **First-author papers**: 626 citations, h-index 9. * advised student.

First-Author Publications

1. **Zhang, S.**, Ma, X.*, Huang, J. *et al.* Thermal Response of Dust and Gas to Shadows in Protoplanetary Disks I: A Beta-Cooling Framework, submitted to *ApJ* (Aug. 2026).
2. **Zhang, S.**, Zhu, Z. & Fairbairn, C. W. Shadow-induced Warps in Protoplanetary Disks. *ApJL* **995**, L33. arXiv: 2511.11358 [[astro-ph.EP](#)] (Dec. 2025).
3. **Zhang, S.** & Zhu, Z. 3D Radiation-hydrodynamical Simulations of Shadows on Transition Disks. *ApJL* **974**, L38. arXiv: 2409.08373 [[astro-ph.EP](#)] (Oct. 2024).
4. **Zhang, S.**, Zhu, Z. & Jiang, Y.-F. Thermal Structure Determines Kinematics: Vertical Shear Instability in Stellar Irradiated Protoplanetary Disks. *ApJ* **968**, 29. arXiv: 2404.05608 [[astro-ph.EP](#)] (June 2024).
5. **Zhang, S.**, Kalscheur, M., Long, F. *et al.* Substructures in Compact Disks of the Taurus Star-forming Region. *ApJ* **952**, 108. arXiv: 2305.03862 [[astro-ph.EP](#)] (Aug. 2023).

6. **Zhang, S.**, Zhu, Z., Ueda, T. *et al.* Porous Dust Particles in Protoplanetary Disks: Application to the HL Tau Disk. *ApJ* **953**, 96. arXiv: 2306.00158 [astro-ph.EP] (Aug. 2023).
7. **Zhang, S.**, Zhu, Z. & Kang, M. PGNets: planet mass prediction using convolutional neural networks for radio continuum observations of protoplanetary discs. *MNRAS* **510**, 4473–4484. arXiv: 2111.15196 [astro-ph.EP] (Mar. 2022).
8. **Zhang, S.**, Hu, X., Zhu, Z. *et al.* Self-consistent Ring Model in Protoplanetary Disks: Temperature Dips and Substructure Formation. *ApJ* **923**, 70. arXiv: 2110.00858 [astro-ph.EP] (Dec. 2021).
9. **Zhang, S.** & Zhu, Z. The effects of disc self-gravity and radiative cooling on the formation of gaps and spirals by young planets. *MNRAS* **493**, 2287–2305. arXiv: 1911.01530 [astro-ph.EP] (Apr. 2020).
10. **Zhang, S.**, Zhu, Z., Huang, J. *et al.* The Disk Substructures at High Angular Resolution Project (DSHARP). VII. The Planet-Disk Interactions Interpretation. *ApJL* **869**, L47. arXiv: 1812.04045 [astro-ph.EP] (Dec. 2018).
11. **Zhang, S.**, Hartmann, L., Zamora-Avilés, M. *et al.* On estimating angular momenta of infalling protostellar cores from observations. *MNRAS* **480**, 5495–5503. arXiv: 1808.04802 [astro-ph.GA] (Nov. 2018).

Contributing-Author Publications

12. Pfeil, T., Ziampras, A., **Zhang, S.** *et al.* Just a Phase? Weakening Vertical Shear Instability Explains Class II Disk Morphologies: Simulations with Dust Coagulation, Sedimentation, Thermal Relaxation, and Backreaction, submitted to ApJL (Aug. 2026).
13. Baronett, S. A.^{*}, Jiang, Y.-F., Zhu, Z., **Zhang, S.** *et al.* A Framework to Model Stellar Irradiated Disks with Frequency-dependent Absorption and Scattering Opacities in Athena++. *arXiv e-prints*. arXiv: 2606.08859 [astro-ph.EP] (June 2026).
14. Jiang, H., Long, F., Macías, E., ... **Zhang, S.** *et al.* Puffed-up Inner Rings and Razor-thin Outer Rings in Structured Protoplanetary Disks. *arXiv e-prints*. arXiv: 2509.13122 [astro-ph.EP] (Sept. 2025).
15. Li, J., Close, L. M., Long, F., ... **Zhang, S.** *et al.* Discovery of H α Emission from a Protoplanet Candidate around the Young Star 2MASS J16120668–3010270 with MagAO-X. *ApJL* **990**, L70. arXiv: 2508.11155 [astro-ph.EP] (Sept. 2025).
16. Uyama, T., Ricci, L., Ygouf, M., ... **Zhang, S.** *et al.* JWST/NIRCam Coronagraphic Search for Hidden Planets in the HD 163296 Protoplanetary Disk. *AJ* **169**, 287. arXiv: 2503.20132 [astro-ph.EP] (June 2025).
17. Zhu, Z., **Zhang, S.** & Johnson, T. M. Asymmetric Temperature Variations In Protoplanetary Disks. I. Linear Theory, Corotating Spirals, and Ring Formation. *ApJ* **980**, 259. arXiv: 2412.09571 [astro-ph.EP] (Feb. 2025).
18. Huang, J., Ansdell, M., Birnstiel, T., ... **Zhang, S.** *et al.* High-resolution ALMA Observations of Richly Structured Protoplanetary Disks in σ Orionis. *ApJ* **976**, 132. arXiv: 2410.03823 [astro-ph.EP] (Nov. 2024).
19. Wallack, N. L., Ruffio, J.-B., Ruane, G., ... **Zhang, S.** *et al.* A Survey of Protoplanetary Disks Using the Keck/NIRC2 Vortex Coronagraph. *AJ* **168**, 78. arXiv: 2408.04048 [astro-ph.EP] (Aug. 2024).

20. Long, F., Andrews, S. M., **Zhang, S.** *et al.* ALMA Detection of Dust Trapping around Lagrangian Points in the LkCa 15 Disk. *ApJL* **937**, L1. arXiv: 2209.05535 [astro-ph.EP] (Sept. 2022).
21. Burrill, B. P., Ricci, L., Harter, S. K.*, **Zhang, S.** *et al.* Investigating the Future Potential of an Upgraded ALMA to Image Planet-forming Disks at Sub-astronomical-unit Scales. *ApJ* **928**, 40. arXiv: 2202.08348 [astro-ph.EP] (Mar. 2022).
22. Andrews, S. M., Elder, W., **Zhang, S.** *et al.* Limits on Millimeter Continuum Emission from Circumplanetary Material in the DSHARP Disks. *ApJ* **916**, 51. arXiv: 2105.08821 [astro-ph.EP] (July 2021).
23. Ueda, T., Kataoka, A., **Zhang, S.** *et al.* Impact of Differential Dust Settling on the SED and Polarization: Application to the Inner Region of the HL Tau Disk. *ApJ* **913**, 117. arXiv: 2104.05927 [astro-ph.EP] (June 2021).
24. Jorquera, S., Pérez, L. M., Chauvin, G., ... **Zhang, S.** *et al.* A Search for Companions via Direct Imaging in the DSHARP Planet-forming Disks. *AJ* **161**, 146. arXiv: 2012.10464 [astro-ph.EP] (Mar. 2021).
25. Harter, S. K.*, Ricci, L., **Zhang, S.** *et al.* Imaging the Dusty Substructures due to Terrestrial Planets in Planet-forming Disks with ALMA and the Next-generation Very Large Array. *ApJ* **905**, 24. arXiv: 2011.08279 [astro-ph.EP] (Dec. 2020).
26. Huang, J., Andrews, S. M., Dullemond, C. P., ... **Zhang, S.** A Multifrequency ALMA Characterization of Substructures in the GM Aur Protoplanetary Disk. *ApJ* **891**, 48. arXiv: 2001.11040 [astro-ph.EP] (Mar. 2020).
27. Zhu, Z., **Zhang, S.**, Jiang, Y.-F. *et al.* One Solution to the Mass Budget Problem for Planet Formation: Optically Thick Disks with Dust Scattering. *ApJL* **877**, L18. arXiv: 1904.02127 [astro-ph.EP] (June 2019).
28. Isella, A., Huang, J., Andrews, S. M., ... **Zhang, S.** *et al.* The Disk Substructures at High Angular Resolution Project (DSHARP). IX. A High-definition Study of the HD 163296 Planet-forming Disk. *ApJL* **869**, L49. arXiv: 1812.04047 [astro-ph.SR] (Dec. 2018).
29. Dullemond, C. P., Birnstiel, T., Huang, J., ... **Zhang, S.** *et al.* The Disk Substructures at High Angular Resolution Project (DSHARP). VI. Dust Trapping in Thin-ringed Protoplanetary Disks. *ApJL* **869**, L46. arXiv: 1812.04044 [astro-ph.EP] (Dec. 2018).
30. Pérez, L. M., Benisty, M., Andrews, S. M., ... **Zhang, S.** *et al.* The Disk Substructures at High Angular Resolution Project (DSHARP). X. Multiple Rings, a Misaligned Inner Disk, and a Bright Arc in the Disk around the T Tauri star HD 143006. *ApJL* **869**, L50. arXiv: 1812.04049 [astro-ph.SR] (Dec. 2018).
31. Kurtovic, N. T., Pérez, L. M., Benisty, M., ... **Zhang, S.** *et al.* The Disk Substructures at High Angular Resolution Project (DSHARP). IV. Characterizing Substructures and Interactions in Disks around Multiple Star Systems. *ApJL* **869**, L44. arXiv: 1812.04536 [astro-ph.SR] (Dec. 2018).
32. Guzmán, V. V., Huang, J., Andrews, S. M., ... **Zhang, S.** *et al.* The Disk Substructures at High Angular Resolution Program (DSHARP). VIII. The Rich Ringed Substructures in the AS 209 Disk. *ApJL* **869**, L48. arXiv: 1812.04046 [astro-ph.SR] (Dec. 2018).

33. Andrews, S. M., Huang, J., Pérez, L. M., ... **Zhang, S. et al.** The Disk Substructures at High Angular Resolution Project (DSHARP). I. Motivation, Sample, Calibration, and Overview. *ApJL* **869**, L41. arXiv: 1812.04040 [astro-ph.SR] (Dec. 2018).
34. Huang, J., Andrews, S. M., Dullemond, C. P., ... **Zhang, S. et al.** The Disk Substructures at High Angular Resolution Project (DSHARP). II. Characteristics of Annular Substructures. *ApJL* **869**, L42. arXiv: 1812.04041 [astro-ph.EP] (Dec. 2018).
35. Huang, J., Andrews, S. M., Pérez, L. M., ... **Zhang, S.** The Disk Substructures at High Angular Resolution Project (DSHARP). III. Spiral Structures in the Millimeter Continuum of the Elias 27, IM Lup, and WaOph 6 Disks. *ApJL* **869**, L43. arXiv: 1812.04193 [astro-ph.SR] (Dec. 2018).
36. Birnstiel, T., Dullemond, C. P., Zhu, Z., ... **Zhang, S.** The Disk Substructures at High Angular Resolution Project (DSHARP). V. Interpreting ALMA Maps of Protoplanetary Disks in Terms of a Dust Model. *ApJL* **869**, L45. arXiv: 1812.04043 [astro-ph.SR] (Dec. 2018).
37. Li, J.-T., Bregman, J. N., Wang, Q. D., ... **Zhang, S.** The Circum-Galactic Medium of Massive Spirals. II. Probing the Nature of Hot Gaseous Halo around the Most Massive Isolated Spiral Galaxies. *ApJS* **233**, 20. arXiv: 1710.07355 [astro-ph.GA] (Dec. 2017).

INVITED TALKS

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| • DAMTP Astrophysics Seminar, University of Cambridge | Cambridge, UK, Nov 2026 |
| • University of Minnesota Duluth Physics and Astronomy Colloquium | Duluth, MN, Sep 2026 |
| • Exploring Planetary Systems in the Era of Time-Domain Astronomy | Honolulu, HI, Jun 2026 |
| • Machine Learning for Disks | Milan, Italy, Apr 2026 |
| • Hydrodynamic and Dusty Turbulence in Protoplanetary Disks | New York, NY, Feb 2026 |
| • University of Illinois Astrophysics Colloquium | Urbana-Champaign, IL, Feb 2026 |
| • Berkeley TAC Seminar | Berkeley, CA, Nov 2025 |
| • Carnegie EPL Astronomy Seminar | Washington, DC, Sep 2024 |
| • University of Chicago Geophysical Sciences Seminar | Chicago, IL, Dec 2023 |
| • TCAN Meeting | Tucson, AZ, Nov 2023 |
| • UW–Madison Monday Science Seminar | Madison, WI, Oct 2023 |
| • University of Hawai'i SPLAT Seminar | Honolulu, HI, Sep 2023 |
| • Star and Planet Formation Seminar, University of Michigan | online, Feb 2022 |
| • Caltech Direct Imaging Group Meeting | online, Dec 2021 |
| • Star Formation: From Clouds to Discs | Malahide, Ireland, Oct 2021 |
| • Peking University KIAA Star and Planet Formation Group Meeting | Beijing, China, Dec 2018 |

SELECTED CONTRIBUTED TALKS AND SEMINARS

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| • Discs on the Exe | Exeter, UK, Jul 2026 |
| • Planet Formation across Various Environments and Epochs | Garching, Germany, May 2026 |
| • 2025 NHFP Symposium | Baltimore, MD, Oct 2025 |

- New York Area Exoplanets Meeting New York, NY, May 2025
- KITP Conference: Planets on Edge Santa Barbara, CA, Apr 2025
- 2024 NHFP Symposium Pasadena, CA, Sep 2024
- 50 Years of Binaries and Discs: Lubow@75 Las Vegas, NV, May 2024
- Princeton Thunch Princeton, NJ, Oct 2023
- NRAO TUNA Talk Charlottesville, VA, Oct 2023
- CfA SMA Seminar Cambridge, MA, Sep 2023
- Harvard ITC Luncheon Cambridge, MA, Sep 2023
- Origins Seminar Tucson, AZ, Sep 2023
- Emerging Researchers in Exoplanet Science (ERES) New Haven, CT, Jun 2023
- Athena++ Workshop New York, NY, May 2023
- Planet Formation Group Meeting, University of Victoria online, Feb 2022
- New Paradigms for Radiatively Efficient Accretion Disks New York, NY, Dec 2021
- Five Years after HL Tau: A New Era in Planet Formation online, Dec 2020
- New Horizons in Planetary Systems Victoria, BC, Canada, May 2019
- 233rd AAS Meeting Seattle, WA, Jan 2019

TEACHING

- **Guest Lecturer**, Graduate Seminar, Columbia University Spring 2025
- **Guest Lecturer**, Computational Physics, UNLV Spring 2024
- **Lecturer & Grader**, ASTR 105: Introductory Astronomy Laboratory, UNLV Spring 2020
- **Lecturer & Grader**, PHYS 151L: General Physics I, UNLV Spring 2019

MENTORING AND ADVISING

- **Stanley Baronett** — Ph.D. student, UNLV (now Postdoc, University of Virginia)
On multi-band radiation-hydrodynamics in protoplanetary disks. Extending the frequency-integrated radiation transport I developed to the multi-frequency thermodynamics of protoplanetary disks. Co-advised with Prof. Zhaohuan Zhu, Dr. Yan-Fei Jiang, and Prof. Phil Armitage; resulting in Baronett et al. (2026).
- **Xiaoyi Ma** — Ph.D. student, KIAA, Peking University
Thermal response to shadows in protoplanetary disks: the role of competing timescales. Modeling the thermal response of the disk in millimeter dust continuum and gas line emission to shadowing in near-IR scattered light; second author on a paper submitted to ApJ.
- **Bronco Yang** — Undergraduate, Columbia University (now Ph.D. student, Rice University)
Roles of planet accretion heating on gap opening. Using Athena++ to study planet-disk interactions including heating by planet accretion in 2D locally isothermal disks.

- **Sarah Harter** — Undergraduate, CSUN (now Ph.D. student, University of Rochester)
Imaging the dusty substructures due to terrestrial planets in planet-forming disks with ALMA and the next-generation Very Large Array. Co-advised with Prof. Luca Ricci; published in ApJ.
- **Fiona Han** — Undergraduate, University of Michigan
Producing synthetic observations of protostellar cores using global simulations. Co-advised with Prof. Lee Hartmann; presented at the University of Michigan Astronomy Undergraduate Poster Session.

PROFESSIONAL SERVICE

- **Referee**, ApJ, ApJL, MNRAS, and PASJ (13 manuscripts) 2019–Present
- **Panelist**, NASA FINESST-25 Review Panel 2026
- **Panelist**, NSF Astronomy and Astrophysics Grants (AAG) Review Panel 2025
- **Scientific Organizing Committee**, 2024 NHFP Symposium, Caltech Sep 2024
- **Mentor**, AMP-UP (Mentorship Program for Upcoming Postdocs) 3 mentees, 2024–2027
- **Sole Organizer**, UNLV Astro Coffee and Astro Journal Club 2019–2020

OUTREACH AND PUBLIC ENGAGEMENT

- **Instructor**, STAR Space School (high school students) Sep 2026
Joint session on exoplanets and planet formation
- **Co-founder, Speaker, and Webmaster**, Astronomy on Tap, Las Vegas 2018–2024
- **Judge**, Beal Bank Science Fair Mar 2022, 2023, 2024
- **Visualization Specialist**, Beijing Planetarium Sep 2020–Mar 2021
Rendered simulations for planetarium shows
- **Speaker**, Public Science Seminar, Beijing Planetarium Feb 2021
“Mars exploration and planet formation” (in Chinese, ~200 general-public attendees)
- **Author**, Amateur Astronomer Magazine (in Chinese) Oct 2020
“GW Ori: ALMA observation of an interesting three-body system”
- **Presenter**, UNLV Art in Science Exhibition Jan 2020
- **Sole Organizer**, Lunar Eclipse on the Strip, Las Vegas Jan 2019
- **Participant**, AAS Astronomy Ambassador Program Jan 2019

SELECTED PRESS COVERAGE

- ALMA Reveals Planets Can Form Under Harsh Radiation [NRAO News](#)
- Ph.D. Candidate Turns Childhood Love of Night Sky Into Studying Disks [UNLV News](#)
- It’s a Planet: New Evidence of Baby Planet in the Making [CfA News](#)
- The Birth of Worlds [Scientific American](#)
- The Epoch of Planet Formation, Times Twenty [NRAO News](#)